

Group A Tool Testing Report

Week 4 : Tools and Function Calling

Project: University Student-Support Case Agent

1. Work Completed

- Ran all 10 planned failure/authorization tests against Oscar's two tools (lookup_case_or_timetable, create_draft_support_case) via tests/run_week4_tool_tests.py, covering normal success, missing parameters, unauthorized access, an unavailable service, an out-of-scope category, and the human-approval gate.
- All 10 execution traces captured in evidence/week4_traces/T01–T10.json, both before and after the fix, so the failure is documented with real evidence rather than just described.

```
tests > week4_tool_tests.json > ...
1  {
2    "id": "T01", "name": "Missing parameter", "tool": "create_draft_support_case",
3    "args": {"student_id": "2300701330", "name": "Opiyo Oscar", "category": "retake", "description": "", "approved": true},
4    "expect_error": "MISSING_FIELD"},
5
6  {"id": "T02", "name": "Unauthorized case lookup (wrong owner)", "tool": "lookup_case_or_timetable",
7   "args": {"student_id": "2300701330", "query_type": "case_status", "case_id": "CASE-0001"},
8   "expect_error": "UNAUTHORIZED"},
9
10 {"id": "T03", "name": "Unavailable service (missing data file)", "tool": "lookup_case_or_timetable",
11  "args": {"student_id": "2300700445", "query_type": "case_status", "case_id": "CASE-0001", "__simulate_outage": true},
12  "expect_error": "SERVICE_UNAVAILABLE"},
13
14 {"id": "T04", "name": "Not found case", "tool": "lookup_case_or_timetable",
15  "args": {"student_id": "2300700445", "query_type": "case_status", "case_id": "CASE-9999"},
16  "expect_error": "NOT_FOUND"},
17
18 {"id": "T05", "name": "Malformed student ID", "tool": "lookup_case_or_timetable",
19  "args": {"student_id": "12345", "query_type": "timetable"},
20  "expect_error": "INVALID_ID"},
21
22 {"id": "T06", "name": "Out-of-scope category (fee wording) not silently accepted", "tool": "create_draft_support_case",
23  "args": {"student_id": "2300701330", "name": "Opiyo Oscar", "category": "other", "description": "Please approve my fee waiver for this semester"},
24  "expect_error": "INVALID_CATEGORY"},
25
26 {"id": "T07", "name": "Approval gate blocks unapproved write", "tool": "create_draft_support_case",
27  "args": {"student_id": "2300701330", "name": "Opiyo Oscar", "category": "retake", "description": "Need to retake BIT2101.", "approved": false},
28  "expect_error": "APPROVAL_DECLINED"},
29
```

```

30 {"id": "T08", "name": "Valid case-status lookup (control case, should succeed)", "tool": "lookup_case_or_timetable",
31 "args": {"student_id": "2300700445", "query_type": "case_status", "case_id": "CASE-0001"},
32 "expect_status": "ok"},
33
34 {"id": "T09", "name": "Valid approved case creation (control case, should succeed)", "tool": "create_draft_support_case",
35 "args": {"student_id": "2300701330", "name": "Opiyo Oscar", "category": "retake", "description": "Need to retake BIT2101.", "approved": true},
36 "expect_status": "draft_created"},
37
38 {"id": "T10", "name": "Invalid query_type value", "tool": "lookup_case_or_timetable",
39 "args": {"student_id": "2300700445", "query_type": "grades"},
40 "expect_error": "INVALID_ID"}
41 ]
42

```

2. Test Results

```

• _tool_tests.pyEAR 4\EMERGING TRENDS\CAPSTONE PROJECT\group_A_ai_agentic_capstone_assignment>
ID      Test                                          Verdict
-----
T01 Missing parameter                               PASS
T02 Unauthorized case lookup (wrong owner)          PASS
T03 Unavailable service (missing data file)         PASS
T04 Not found case                                 PASS
T05 Malformed student ID                           PASS
T06 Out-of-scope category (fee wording) not silently acceptedPASS
T07 Approval gate blocks unapproved write          PASS
T08 Valid case-status lookup (control case, should succeed)PASS
T09 Valid approved case creation (control case, should succeed)PASS
T10 Invalid query_type value                         PASS
-----
10/10 passed. week4_traces in evidence/week4_traces/
❖ (venv) PS D:\YEAR 4\EMERGING TRENDS\CAPSTONE PROJECT\group_A_ai_agentic_capstone_assignment>

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